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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

Representation of Continuous Time Signal Fourier Series

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

21	16	1992 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	13 questions
Marks 2	8 questions
Answer key	at the end

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Marks 1 13 questions

Q1. GATE ECE 2023

In the table shown below, match the signal type with its spectral characteristics.

	Signal type		Spectral characteristics
(i)	Continuous, aperiodic	(a)	Continuous, aperiodic
(ii)	Continuous, periodic	(b)	Continuous, periodic
(iii)	Discrete, aperiodic	(c)	Discrete, aperiodic
(iv)	Discrete, periodic	(d)	Discrete, periodic

Open solution

(A) $(i) \rightarrow (a), (ii) \rightarrow (b), (iii) \rightarrow (c), (iv) \rightarrow (d)$

(B) $(i) \rightarrow (a), (ii) \rightarrow (c), (iii) \rightarrow (b), (iv) \rightarrow (d)$

(C) $(i) \rightarrow (d), (ii) \rightarrow (b), (iii) \rightarrow (c), (iv) \rightarrow (a)$

(D) $(i) \rightarrow (a), (ii) \rightarrow (c), (iii) \rightarrow (d), (iv) \rightarrow (b)$

Q2. GATE ECE 2018

Let the input be u and the output be y of a system, and the other parameters are real constants. Identify which among the following systems is not a linear system:

Open solution

(A) $\frac{d^3y}{dt^3} + a_1\frac{d^2y}{dt^2} + a_2\frac{dy}{dt} + a_3y = b_3u + b_2\frac{du}{dt} + b_1\frac{d^2u}{dt^2}$ (with initial rest conditions)

(B) $y(t) = \int_0^t e^{\alpha(t-\tau)}\beta u(\tau)d\tau$

(C) y = au + b, $b \neq 0$

(D) y = au

Q3. GATE ECE 2018

Let $x(t)$ be a periodic function with period T = 10. The Fourier series coefficients for this series are denoted by a_k, that is

Open solution

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t}$$

The same function $x(t)$ can also be considered as a periodic function with period T' = 40. Let b_k be the Fourier series coefficients when period is taken as T'. If $\sum_{k=-\infty}^{\infty} |a_k| = 16$, then $\sum_{k=-\infty}^{\infty} |b_k|$ is equal to

(A) 256 (B) 64 (C) 16 (D) 4

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Q4. GATE ECE 2017 Set 1

Open solution

A periodic signal x(t) has a trigonometric Fourier series expansion

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos\ n\omega_0 t + b_n \sin\ n\omega_0 t)$$

If $x(t) = -x(-t) = -x(t - \pi/\omega_0)$, we can conclude that

(A) a_n are zero for all n and b_n are zero for n even

(B) a_n are zero for all n and b_n are zero for n odd

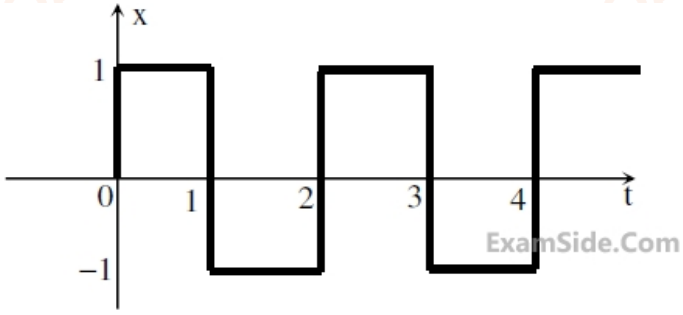
(C) a_n are zero for n even and b_n are zero for n odd

(D) a_n are zero for n odd and b_n are zero for n even

Q5. GATE ECE 2014 Set 2

Open solution

Consider the periodic square wave in the figure shown.

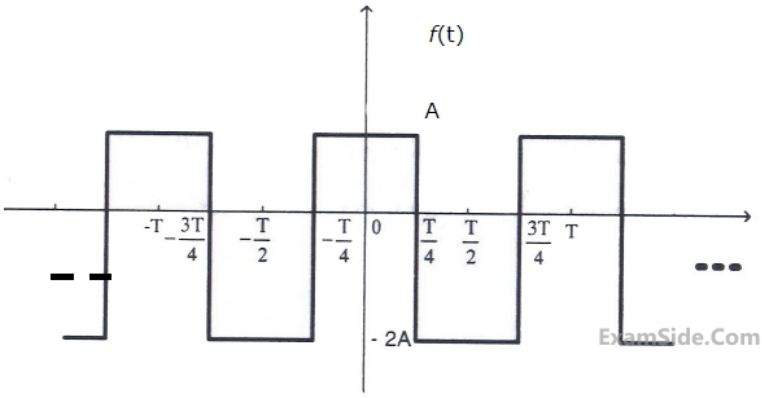


The ratio of the power in the 7th harmonic to the power in the 5th harmonic for this waveform is closest in value to _____.

Q6. GATE ECE 2010

Open solution

The trigonometric Fourier series for the waveform f(t) shown below contains



(A) only cosine terms and zero value for the dc component

(B) only cosine terms and a positive value for the dc component

(C) only cosine terms and a negative value for the dc component

(D) only sine terms and a negative for the dc component

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Q7. GATE ECE 2009

The Fourier series of a real periodic function has only
P. Cosine terms if it is even
Q. Sine terms if it is even
R. Cosine terms if it odd
S. Sine terms if it is odd
Which of the above statement are correct?

(A) P and S (B) P and R (C) Q and S (D) Q and R

Open solution

Q8. GATE ECE 2009

A function is given by f(t) = $\sin^2 t$ +cos2t . Which of the following is true?

(A) f has frequency components at 0 and $1/2\pi$ Hz
(B) f has frequency components at 0 and $1/\pi$ Hz
(C) f has frequency components at $1/2\pi$ and $1/\pi$ Hz
(D) f has frequency components at 0, $1/2\pi$ and $1/\pi$ Hz

Open solution

Q9. GATE ECE 2005

Choose the function f (t);$-\infty$ < 1 < $+\infty$, for which a Fourier series cannot be defined.

(A) 3sin(25t) (B) 4cos(20t+3)+2sin(10t)
(C) exp (−|t|) sin(25t) (D) 1

Open solution

Q10. GATE ECE 2002

Which of the following cannot be the Fourier series expansion of a periodic signal?

(A) x(t) = 2cos t + 3 cos 3t (B) x(t) = 2cos πt + 7 cos t
(C) x(t) = cos t + 0.5 (D) x(t) = 2cos 1.5πt + sin 3.5 πt

Open solution

Q11. GATE ECE 1998

The trigonometric Fourier series of a periodic time function can have only

(A) cosine terms (B) sine terms
(C) cosine and sine terms (D) d.c. and cosine terms

Open solution

Q12. GATE ECE 1996

The trigonometric Fourier series of an even function of time does not have the

(A) dc term (B) cosine terms
(C) sine terms (D) odd harmonic terms

Open solution

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Q13. GATE ECE 1994

The Fourier Series of an odd periodic function, contains only

(A) odd harmonics (B) even harmonics (C) cosine terms (D) sine terms

Open solution

Marks 2 8 questions

Q14. GATE ECE 2026

The continuous time signal $x(t)$ is real, periodic with period T and satisfies the Dirichlet conditions.

The Fourier series representation of $x(t) = \sum_{n=-\infty}^{\infty} a_n e^{j\left(\frac{2\pi nt}{T}\right)}$ and $x(t)$ satisfies the following:

$x\left(t - \frac{T}{2}\right) = -x(t)$

For any integer m, which of the following options is correct?

(A) $a_{2m} = 0$ (B) $a_{2m} = 1$ (C) $a_{2m} = a_{2m+1}$ (D) $a_{2m} = -1$

Open solution

Q15. GATE ECE 2026

Let $x_1(t) = \cos(2\pi nt)$ and $x_2(t) = 2\sin(4\pi nt)$ represent two sinusoids for a positive integer n and $-\infty < t < \infty$.

Which of the following statements about $x_1(t)$ and $x_2(t)$ is/are valid?

(A) $x_1(t)$ and $x_2(t)$ are orthogonal to each other over $0 \le t < 1/n$.

(B) $x_1(t)$ and $x_2(t)$ are orthonormal to each other over $0 \le t < 1/n$.

(C) $x_2(t)$ is a harmonic of $x_1(t)$.

(D) $x_1(t)$ and $x_2(t)$ are non-orthogonal to each other over $0 \le t < 1/(2n)$.

Open solution

Q16. GATE ECE 2025

Let $f(t)$ be a periodic signal with fundamental period $T_0 > 0$. Consider the signal $y(t) = f(\alpha t)$, where $\alpha > 1$.

The Fourier series expansions of $f(t)$ and $y(t)$ are given by

$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\frac{2\pi}{T_0}kT}$ and $y(t) = \sum_{k=-\infty}^{\infty} d_k e^{j\frac{2\pi}{T_0}\alpha kT}$.

Which of the following statements is/are TRUE?

(A) $c_k = d_k$ for all k

(B) $y(t)$ is periodic with a fundamental period αT_0

(C) $c_k = d_k/\alpha$ for all k

(D) $y(t)$ is periodic with a fundamental period T_0/α

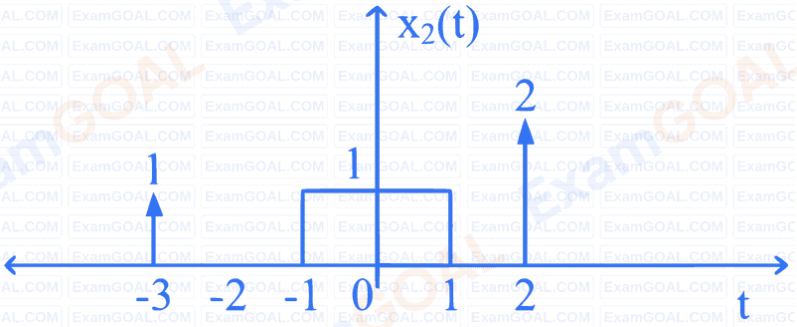
Open solution

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Q17. GATE ECE 2023

Open solution

Let $x_1(t) = u(t + 1.5) - u(t - 1.5)$ and $x_2(t)$ is shown in the figure below. For $y(t) = x_1(t) * x_2(t)$, the $\int_{-\infty}^{\infty} y(t)dt$ is _____ (rounded off to the nearest integer).



Q18. GATE ECE 2021

Open solution

The exponential Fourier series representation of a continu-ous-time periodic signal $X(t)$ is defined as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jkw_0 t}$$

Where ω_0 is the fundamental angular frequency of $x(t)$ and the coefficients of the series are a_k. The following information is given about $x(t)$ and a_k.

I. $x(t)$ is real and even, having a fundamental period of 6

II. The average value of $x(t)$ is 2

III. $a_k = \begin{cases} k, 1 \le k \le 3 \\ 0, k > 3 \end{cases}$

The average power of the signal $x(t)$ (rounded off one decimal place) is ____

Q19. GATE ECE 2017 Set 1

Open solution

Let x(t) be a continuous time periodic signal with fundamental period T = 1 seconds. Let $\{a_k\}$ be the complex Fourier series coefficients of x(t), where k is integer valued. Consider the following statements about x(3t):

I. The complex Fourier series coefficients of x(3t) are $\{a_k\}$ where k is integer valued

II. The complex Fourier series coefficients of x(3t) are $\{3a_k\}$ where k is integer valued

III. The fundamental angular frequency of x(3t) is 6π rad/s

For the three statements above, which one of the following is correct?

(A) only II and III are true
(B) only I and III are true
(C) only III is true
(D) only I is true

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Q20. GATE ECE 2001

The PSD and the power of a signal g(t) are, respectively, $S_g(\omega)$ and P_g. The PSD and the power of the signal ag(t) are, respectively

(A) $a^2S_g(\omega)\ and\ a^2P_g$ (B) $a^2S_g(\omega)\ and\ aP_g$ (C) $aS_g(\omega)\ and\ a^2P_g$ (D) $aS_g(\omega)\ and\ aP_g$

Open solution

Q21. GATE ECE 1992

Which of the following signals is/are periodic?

(A) S (t) = cos2t + cos3t + cos5t

(B) S (t) = exp (j8πt)

(C) S (t) = exp (−7t) sin10πt

(D) S (t) = cos2t cos 4t

Open solution

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Answer Key

Marks 1

1. **B**	2. **C**	3. **C**	4. **A**	5. **0.5**
6. **C**	7. **A**	8. **B**	9. **C**	10. **B**
11. **C**	12. **C**	13. **D**		

Marks 2

14. **A**	15. **A, C, D**	16. **A, D**	17. **15**	18. **32**
19. **B**	20. **A**	21. **A, B, D**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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